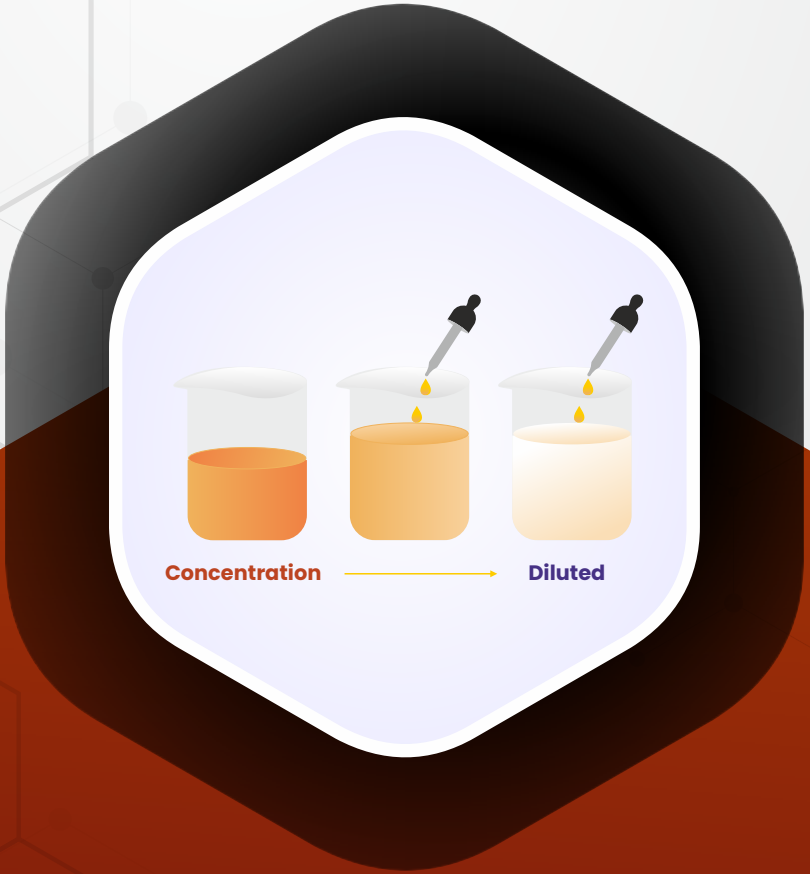
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Concentration Terms

(English Medium)

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CONCENTRATION TERMS

1. SOLUTION

A solution is a homogenous mixture of two or more pure substances whose composition may be altered within certain limits. Though the solution is homogenous in nature, yet it retains the properties of its constituents.

Generally solution is composed of two components, **solute** and **solvent**. Such type of solution is known as binary solution.

Solvent is that component in solution whose physical state is the same as that of the resulting solution while other component is called as solute. If the physical state of both component is same, then the component in excess by moles is known as solvent and other one as solute. A solution may exist in any physical state such as liquid, solid and gaseous state.

Types of Solutions

Type of Solutions	Solute	Solvent	Common Example
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

2. CONCENTRATION OR STRENGTH OF SOLUTION :

The concentration of a solution is the amount of solute dissolved in a known amount of the solvent or solution. Solution can be described as dilute or concentrated solution as per their concentration. A dilute solution has a very small quantity of solute while concentrated solution has a large quantity of solute in solution. Various concentration terms are as follows:

2.1 Mass percentage :

It may be defined as the number of parts of mass of solute per hundred parts by mass of solution.

$$\textbf{\% by mass}\left(\frac{w}{W}\right) = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100$$

[X % by mass means 100 gm solution contains X gm solute and hence (100 – X) gm solvent]

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2.2 Mass-volume percentage (W/V %) :

It may be defined as the mass of solute (in gm) present in 100 cm^3 of solution. For example, If 100 cm^3 of solution contains 5 g of sodium hydroxide, then the mass-volume percentage will be 5% solution.

$$\% \left(\frac{w}{V}\right) = \frac{\text{wt. of solute}}{\text{volume of solution}} \times 100$$

[X % $\left(\frac{w}{V}\right)$ means 100 ml solution contains X gm solute]

2.3 Volume Percent :

It can be represented as % v/v or % volume and normally used for the solutions in which both components are in liquids state. It is the number of parts of by volume solute per hundred parts by volume of solution. Therefore,

$$\% \left(\frac{v}{V}\right) = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100$$

2.4 $\textbf{Mole \%} = \frac{\text{Moles of solute}}{\text{Total moles}} \times 100$

- For gases, **% by volume** is same as **mole %**

2.5 Mole Fraction (X) :

Mole fraction may be defined as the ratio of number of moles of one component to the total number of moles of all the components (solute and solvent) present in solution. It is denoted by letter X and the sum of all mole fractions in a solution is always equals to one.

$$\text{Mole fraction (X)} = \frac{\text{Moles of solute}}{\text{Total moles}}$$

Mole fraction does not depend upon temperature and can be extended to solutions having more than two components.

2.6 Molarity (M) :

Molarity is most common unit for concentration of solution. It is defined as the number of moles of solute present in one litre or one dm^3 of the solution or millimol of solute present in one mL of solution.

$$\text{Molarity (M)} = \frac{\text{Mole of solute}}{\text{volume of solution in litre}}$$

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2.7 **Molality (m) :** The number of mole of the solute present in 1000 g of the solvent is known as molality of solution. It represented by letter 'm'.

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

The unit of molality is mol/kg and is not affected by temperature.

2.8 **Parts per million (ppm) :** The very low concentration of solute in solution can be expressed in ppm. It is the number of parts by mass of solute per million parts by mass of the solution.

$$\text{Parts per million (ppm)} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

◆ Get yourselves very much comfortable in their inter conversion. It is very handy.

Concentration Type	Mathematical Formula	Concept
Percentage by mass	$\%\left(\frac{w}{w}\right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$	Mass (in gm) of solute present in 100 gm of solution.
Volume percentage	$\%\left(\frac{v}{v}\right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$	Volume (cm^3) of solute present in 100 cm^3 of solution.
Mass-volume percentage	$\%\left(\frac{w}{v}\right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$	Mass (in gm) of solute present in 100 cm^3 of solution.
Parts per million	$\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$	Parts by mass of solute per million parts by mass of the solution
Mole fraction	$X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + ...}$ $X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + ...}$	Ratio of number of moles of one component to the total number of moles.
Molarity	$M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$	Moles of solute in one litre of solution.
Molality	$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$	Moles of solute in one kg of solvent

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Ex.1 ***Calculate the mole fractions of the components of the solution composed by 92 g glycerol and 90 g water ? (M (water) = 18 ; M (glycerol) = 92)***

Sol. Moles of water = 90 g / 18 g = 5 mol water

Moles of glycerol = 92 g / 92 g = 1 mol glycerol

Total moles in solution = 5 + 1 = 6 mol

Mole fraction of water = 5 mol / 6 mol = 0.833

Mole fraction of glycerol = 1 mol / 6 mol = 0.167

Ex.2 ***What will be the molarity of solution when water is added to 16.4g $Ca(NO_3)_2$ to make 100 mL of solution?***

Sol. Mol of $Ca(NO_3)_2$ = 16.4/ 164 = 0.1

Molarity = Mole of solute / Volume of solution (L) = 0.10 mol / 0.10 L

Therefore, Molarity of given solution = 1.0 M

Ex.3 ***Calculate the molality of a solution containing 20 g of sodium hydroxide (NaOH) in 250 g of water?***

Sol. Moles of sodium hydroxide = 20 / 40 = 0.5 mol NaOH

250 gm = 0.25 kg of water

Hence molality of solution = Mole of solute / Mass of solvent (kg)= 0.5 mol / 0.25 kg

or Molality(m) = 2.0 m

Ex.4 ***Calculate the gram of copper sulphate ($CuSO_4$) needed to prepare 250.0 mL of 1.00 M $CuSO_4$?***

Sol. Moles of $CuSO_4 = M \times V = 1 \times \frac{250}{1000}$

Molar mass of copper sulphate = 159.6 g/mol

Hence Mass of copper sulphate (gm) = Moles of $CuSO_4$ × Molar mass of copper sulphate.

$= 1 \times \frac{250}{1000} \times 159.6$ g/mol = 39.9 gm of Copper sulphate

Ex.5 ***How many gram of H_2SO_4 are present in 500 ml of 0.2M H_2SO_4 solution ?***

Sol. $M = \frac{\text{moles}}{\text{vol.}} \Rightarrow$ moles of $H_2SO_4 = M \times V = 0.2 \times \frac{500}{1000} L = 0.1$

Mass of $H_2SO_4 = 0.1 \times 98 = 9.8$ g

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Ex.6 ***Calculate the ppm of mercury in water in a sample containing 30 mg of Hg in 500 ml of solution.***

Sol. $\text{Parts per million} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$

Mass of Hg = 30 mg

Mass of water = 500/1 = 500g = 50×10^4 mg

(density = mass / volume ; density of water 1 g / ml) $w = \frac{v}{d}$

Therefore, ppm of mercury = $\frac{30 \times 10^6}{50 \times 10^4} = 60$ ppm of mercury

Ex.7 ***A 100g NaOH solution has 20g NaOH. Find molality.***

Sol. $m = \frac{20/40}{100-20} \times 1000 = \frac{500}{80} = 6.25\,\text{mol/kg}$

Ex.8 ***Find molality of aqueous solution of CH_3COOH whose molarity is 2M and density d = 1.2 g/mL.***

Sol. Hint : $\frac{\mathbf{1000 \times M}}{\mathbf{1000 \times d - MMs}}$

where d = density in gL^{-1} , M = Molarity, m = molality, M_S = molar mass of solute.

$m = \frac{2}{1200 - 2 \times 60} \times 1000 = 1.85$ m

Ex.9 ***A solution has 80% $\frac{w}{w}$ NaOH with density $2gL^{-1}$. Find (a) Molarity (b) Molality of solution.***

Sol. Let V be vol. of solution, in L

$$\text{Mass of solute} = (d \times V) \times \frac{\left(\%\frac{w}{w}\right)}{100} = 2 \times V \times \frac{80}{100} = 1.6V$$

(a) $M = \frac{1.6V/40}{V} = 0.04$ m (b) $m = \frac{1.6V/40}{2V - 1.6V} \times 1000 = 100\,\text{mol kg}^{-1}$

Ex.10 ***4.450 g sulphuric acid was added to 82.20 g water and the density of the solution was found to be 1.029 g/cc at 25°C and 1 atm pressure. Calculate***

Sol. ***(a) the weight percent,***

(b) the mole fraction ,

(c) the mole percent,

(d) the molality ,

(e) the molarity of sulphuric acid in the solution under these conditions.

Sulphuric acid = 4.450 g , Water = 82.20 g $\Rightarrow$ Wt. of solution = 86.65 g

$\therefore$ Density of solution = 1.029 g/cc.

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(a) $\text{Weight percent} = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100 = \frac{4.450}{86.65} \times 100 = 5.14$

(b) Mole fraction :

$\text{Mole of solute} = \frac{\text{wt. of solute}}{\text{mol wt. of solute}} = \frac{4.45}{98} = 0.0454$

$\text{Mole of solvent} = \frac{82.20}{18} = 4.566$

Total moles in solution = 0.0454 + 4.566 = 4.6114

$\text{Mole fraction of solute} = \frac{0.0454}{4.6114} = 0.0098$

(c) Mole percent = mole fraction of solute × 100 = 0.0098 × 100 = 0.98

(d) $\text{Molality} = \frac{\text{moles of solute}}{\text{mass of solvent (in gm)}} \times 1000$

$= \frac{0.0454 \times 1000}{82.2} = 0.552$

(e) $\text{Molarity} = \frac{\text{moles of solute}}{\text{litre of solution}}$

$\text{Volume of solution} = \frac{\text{Mass}}{\text{Density}} = \frac{86.65}{1.029}\text{ml} = \frac{86.65}{1.029 \times 1000}\text{litre}$

$\text{Molarity} = \frac{0.0454}{\frac{86.54}{1.029 \times 1000}} = \frac{0.0454 \times 1000 \times 1.029}{86.65} = 0.539$

Ex.11 ***Find number of Na^+ & PO_4^{-3} ions in 250 ml of 0.2M Na_3PO_4 solution.***

Sol. $Na_3PO_4 + aq. \longrightarrow 3Na^+(aq) + PO_4^{-3}(aq)$ [Ionic compound when added to water ionize completely].

50 millimoles (m.m.) 150 mm 50 mm

No. of Na^+ ions = $150 \times 10^{-3} \times N_A$; No. of PO_4^{-3} ions = $50 \times 10^{-3} \times N_A$

Ex.12 ***80g NaOH was added to 2L water. Find molality of solution if density of water = 1g/mL***

Sol. $m = \frac{\text{moles of NaOH}}{\text{mass of } H_2O} \times 1000 = \frac{80/40}{2 \times 1000} \times 1000 = 1\,\text{molal}$

Ex.13 The average concentration of Na^+ ion in human body is 3.0 to 3.9 gm per litre. The molarity of Na^+ ion is about.

0.15 M

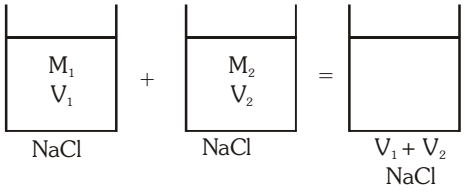
Sol. $M_{Na^+} = \frac{n_{solute}}{\text{volume of solution in Lt}} = \frac{\frac{3+3.9}{2}}{23} = \frac{6.9}{46} = 0.15M$

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3. MIXING OF AND DILUTION SOLUTIONS :

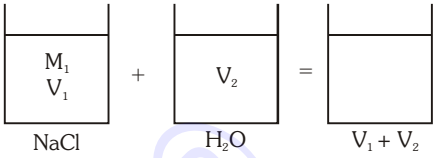
It is based on law of conservation of moles of solute

(i) Two solutions having same solute



$$\text{Final molarity} = \frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

(ii) Dilution Effect : When a solution is diluted, the moles of solute do not change but molarity changes.



$$\text{Final molarity} = \frac{M_1V_1}{V_1 + V_2}$$

Ex.14 ***50 ml 0.2 M H_2SO_4 is mixed with 50 ml 0.3M H_2SO_4. Find molarity of final solution.***

Sol. $M_f = \frac{\text{Total moles of } H_2SO_4}{\text{Total volume}} = \frac{50\times0.2\times10^{-3} + 50\times10^{-3}\times0.3}{(50+50)\times10^{-3}} = 0.25 \text{ M.}$

Ex.15 ***Find final molarity in each case :***

(i) 500 ml 0.1 M HCl + 500 ml 0.2M HCl

(ii) 50 ml, 0.1M HCl + 150 ml, 0.3M HCl + 300 ml H_2O

(iii) 4.9g H_2SO_4 + 250 ml H_2O + 250 ml 0.1 M H_2SO_4

Sol.

(i) $M_f = \frac{500\times0.1 + 500\times0.2}{500+500} = 0.15 \text{ M.}$

(ii) $M_f = \frac{50\times0.1 + 150\times0.3}{50+150+300} = \frac{50}{500} = 0.1 \text{ M}$

(iii) $M_f = \frac{\frac{4.9}{98} + \frac{250}{1000}\times0.1}{\left(\frac{250+250}{1000}\right)} = \frac{50+25}{500} = 0.15 \text{ M}$

Ex.16 ***How much water should be added to 2M HCl solution to form 1 litre of 0.5 M HCl ?***

Sol. Let V be initial volume

Then mol of HCl = constant

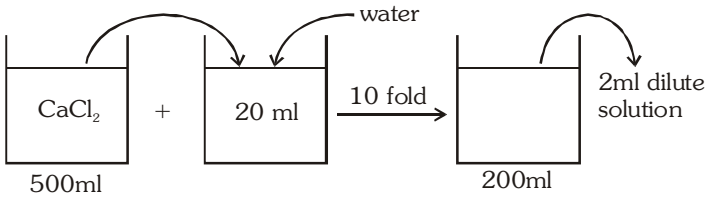
$2 \times V = 1 \times 0.5 \Rightarrow V = 0.25 \text{ L}$

Volume of water added = 1 – 0.25 = 0.75 L

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Ex.17 ***1.11g $CaCl_2$ is added to water forming 500 ml of solution. 20 ml of this solution is taken and diluted 10 folds. Find moles of Cl^- ions in 2 ml of diluted solution.***

Sol. Moles of $CaCl_2 = \frac{1.11}{111} = 0.01$ mol



Moles of $CaCl_2$ in 20ml solution $= \frac{0.01}{500} \times 20 = \frac{0.01}{25}$

In 200 ml solution, moles of $CaCl_2 = \frac{0.01}{25}$ [Dilution does not change moles of solute]

In 2 ml of dilute solution moles of $CaCl_2 = \frac{0.01/25}{200} \times 2 = \frac{0.01}{2500} = 4 \times 10^{-6}$

$\therefore$ moles of $Cl^- = 2 \times 4 \times 10^{-6} = 8 \times 10^{-6}$

Ex.18 ***What volumes of 1M & 2M H_2SO_4 solution are required to produce 2L of 1.75M H_2SO_4 solution?***

Sol. Let XL be vol. of 1M solution.

$\therefore$ (2 – X)L is vol. of 2M solution.

Moles of H_2SO_4 : $2 \times 1.75 = 1(X) + (2 - X)2 \Rightarrow X = 0.5$ L

i.e. 0.5L of 1M & 1.5 L of 2M solution are required.

Ex.19 ***A solution is made by mixing 300 ml 1.5M $Al_2(SO_4)_3$ + 300 ml 2M $CaSO_4$ + 400 ml 3.5M $CaCl_2$***

Find final molarity of (1) SO_4^{-2}, (2) Ca^{2+}, (3) Cl^-. [Assume complete dissociation of these compounds].

Sol. (1) $[SO_4^{-2}]_f = \frac{\text{Total moles}}{\text{Total volume}} = \frac{300 \times 1.5 \times 10^{-3} \times 3 + 300 \times 2 \times 10^{-3}}{(300 + 300 + 400) \times 10^{-3}} = 1.95M$

(2) $[Ca^{+2}]_f = \frac{300 \times 2 + 400 \times 3.5}{1000} = 2M$

(3) $[Cl^-]_f = \frac{400 \times 3.5 \times 2}{1000} = 2.8M$

Ex.20 ***A solution of KCl has a density of 1.69 g mL^{-1} and is 67% by weight. Find the density of the solution if it is diluted so that the percentage by weight of KCl in the diluted solution is 30%.***

Sol. Let the volume of the KCl solution be 100 mL,

Weight of KCl solution = 100 × 1.69 = 169 g

100 g of solution contains = 67 g of KCl

169 g of solution $= \frac{67}{100} \times 169 = 113.23$g

Lex x mL of H_2O be added.

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New volume of solution = (100 + x) mL

New weight of solution = (169 + x) g

(Since x mL of H_2O = x g of H_2O, d_{H_2O} = 1)

New percentage of the solution = 30%

$$\% \text{ by weight} = \frac{\text{weight of solute} \times 100}{\text{weight of solution}}$$

$$30 = \frac{113.23}{(169+x)} \times 100$$

x = 208.43 mL = 208.43 g

$$\text{New density} = \frac{\text{New weight of solution}}{\text{New volume of solution}} = \frac{(169+x)}{(100+x)}$$

$\therefore$ d = 1.224 gm/m*l*

Ex.21 ***Calculate the amount of the water ("in mℓ") which must be added per ml to a given solution of concentration of 40 mg silver nitrate per ml, to yield a solution of concentration of 16 mg silver nitrate per ml ?***

Sol. Before dilution After dilution

(n_{solute}) = (n_{solute})

$M_i\ V_i$ = $M_f\ V_f$

$$\frac{\frac{40}{170}}{1} \times 1 = \frac{\frac{16}{170}}{1}(1+V)$$

$\Rightarrow$ V = 1.5 mL

Ex.22. ***100 ml, 3%(w/v) NaOH solution is mixed with 100 ml, 9%(w/v) NaOH solution. The molarity of final solution is-***

Sol. Total NaOH in 100 ml (1st solution) = 3 gm

Total NaOH in 100 ml (2nd solution) = 9 gm

$$\therefore \text{ Molarity} = \left(\frac{12/40}{200/1000}\right) = 1.5\text{M}$$

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Ex.23. ***1120 gm of 2 'm' urea solution is mixed with 2480 gm of 4 'm' urea solution. Calculate the molality of the resulting solution?***

Sol. Let 2 m, 1120 g solution have mass of solute = w gm

$\therefore$ solvent = (1120 – w) gm

& Let 4 m, 2480 g solution have mass of solute = w' gm

$\therefore$ solvent = (2480 – w) gm

$$\text{molality} = \frac{n_{\text{solute}}}{\text{wt. of solvent in kg}}$$

$$2 = \frac{w/60}{1120 - w} \times 1000$$

$w = 120$ gm

$$\& \quad 4 = \frac{w'/60}{2480 - w'} \times 1000$$

$w' = 480$ gm

$$\text{resulting molality m} = \frac{\frac{120 + 480}{60}}{1120 - 120 + 2480 - 480} \times 1000 = 3.33 \text{ m}$$

4. VOLUME STRENGTH OF H_2O_2 SOLUTION :

It represents the volume of O_2 (in litre) at STP that can be obtained from 1 litre of H_2O_2 solution, when H_2O_2 decomposes as :

$$H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$$

Volume strength is normally represented as 'x vol' or 'x V'.

Volume Strength of H_2O_2 Solution = 11.35 × molarity

Ex.24 ***Find the % w/v of "10 V" H_2O_2 solution-***

Sol. $$\text{Molarity (M) of solution} = \frac{\text{volume strength}}{11.35} = \frac{10}{11.35}$$

$$\% \left(\frac{w}{v}\right) = \frac{M \times \text{mol. wt. of solute}}{10} = \frac{10}{11.35} \times \frac{34}{10} = 3\%$$

Ex.25 ***$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$***

Under conditions where 1 mole of gas occupies 24 dm^3, ***X L of*** $\frac{1}{24}$ M ***solution of H_2O_2 produces 3 dm^3 of O_2. Thus X is :-***

Sol. $$\text{moles of } H_2O_2 = \frac{1}{24} \times X$$

$$\text{moles of } O_2 = \frac{3}{24} = \frac{1}{8}.$$

$$\text{moles of } H_2O_2 = \frac{1}{4} = \frac{X}{24}$$

$X = 6$

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PREVIOUS YEARS SOLVED EXAMPLES

Q.1 **One gm of charcoal absorbs 100 ml 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = 3.01×10^2 m^2/gm. [JEE'2003]**

Ans. $\mathbf{5 \times 10^{-19}\ m^2}$

Sol. Final molarity = 0.5 – 0.49 = 0.01 M

$$\text{mole} = M \times v = 0.01 \times \frac{100}{1000} = 10^{-3}$$

$$\text{no of molecule} = \text{moles} \times N_A = 10^{-3} \times N_A = 6.02 \times 10^{20}$$

1 gm contain charcoal = 3.01×10^2 m^2

6.02×10^{20} molecule of acetic acid absorbed charcoal = 3.01×10^2

1 molecule of acetic acid adsorbed charcoal = 3.01×10^2 m^2

6.02×10^{20} molecule of acetic acid absorbed charcoal = 3.01×10^2

$$\text{1 molecule of acetic acid adsorbed charcoal} = \frac{3.01 \times 10^2}{6.02 \times 10^{20}} = 5 \times 10^{-19}\ m^2$$

Q.2 Calculate the molarity of pure water using its density to be 1000 kg m^{-3}. [JEE'2003]

Ans. $\mathbf{55.5\ mol\ L^{-1}}$

Sol. $$M = \frac{1000}{MW} = \frac{1000}{18} = 55.5$$

Q.3 6.02×10^{21} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is - [AIEEE-2004]

(A) 0.001 M (B) 0.01 M (C) 0.02 M (D) 0.1 M

Ans. **(D)**

Sol. $$\text{Moles of urea} = \frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = \frac{1}{100}$$

$$\text{Molarity} = \frac{1000}{100 \times 100} = 0.1\ M$$

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Important Notes

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